1. **Proposal 1 — Similarity vs Difference**
    a. **How similar or different are music preferences across regions, and how is that changing over time?**
    b. Problem: We don't know whether global music discovery is expanding people's tastes, or just reinforcing the same global hits everywhere.
        i. Are we actually gaining listening diversity regionally? Or are we just repeating the same top songs globally?
    c. Audience:
        i. music listeners/explorers
        ii. students (globalization, culture)
    d. Use cases:
        i. Compare music "taste profiles" across countries
        ii. Identify regions with unique vs globally aligned tastes
        iii. See whether music is becoming more uniform globally
    e. General data ideas:
        i. Spotify charts by country (Top 50/200)
        ii. Spotify audio features:
            1. Danceability
            2. Energy
            3. Valence
            4. Tempo
        iii. genre tags
    f. Features?
        i. Region Comparison Dashboard
            1. Select 2–3 countries
            2. Compare:
                a. average audio features
                b. top artists
                c. genre distribution
        ii. Similarity Over Time Trend
            1. Show how close regions are becoming
            2. Simple metric: distance between audio feature averages
2. **Proposal 2 — Trend Flow (Origin vs Adoption)**
    a. **How does music move between regions, and which regions act as trend originators versus adopters?**

b. Problem: We don't understand who actually drives global music trends, and which regions are underrecognized as cultural originators.
c. Audience:
    i. students / academic angle
    ii. music trend explorers
d. Use cases:
    i. Track where songs appear first vs later
    ii. Identify "trend leader" vs "follower" regions
    iii. Visualize how music spreads geographically
e. General data ideas:
    i. time-series chart data by country
    ii. track/artist appearance dates across regions
    iii. ranking timelines
f. Features?
    i. Spread Timeline View
        1. Select an artist/song
        2. See:
            a. first appearance by region
            b. timeline of spread
    ii. "Origin vs Adoption" Indicator
        1. Label regions as:
            a. early adopters
            b. late adopters

3. Proposal 3 — Overlap + Discovery Gap

a. **How much overlap exists between popular music across regions, and how is that changing over time?**
b. Problem: People are not exposed to music they would likely enjoy because it hasn't crossed into their region yet.
c. Audience:
    i. Music listeners
    ii. Discovery-focused users
d. Use cases:
    i. See what music is globally shared vs region-specific
    ii. Identify "hidden" music outside your region
    iii. Identify "hidden" music outside your region
e. General data ideas:

        i. top charts by country
        ii. artist/song overlap across regions
        iii. % shared vs unique content
    f. <u>Features?</u>
        i. "Hidden Music" Explorer
            1. User selects their country
            2. Shows:
                a. Artists popular elsewhere, but not locally
        ii. Overlap comparison tool
            1. Select 2 countries
            2. See:
                a. % shared songs
                b. Unique artists per region
        iii. Global vs local
            1. Track:
                a. How much music is shared globally over time

DATA SOURCE IDEAS- not fully vetted for practicality, just a big 'ol giant brainstorm

- **Media & Exposure Data**
    - Directly supports "Bad Bunny / K-pop exposure" insight.
    - <u>Data sources:</u>
        - IMDb datasets (movie/TV soundtracks)
        - Spotify soundtrack playlists (if accessible)
        - TMDB API (movie + show metadata)
        - Billboard soundtrack charts
    - <u>Use cases:</u>
        - Track songs/artists appearing in:
            - Movies
            - TV shows
        - Compare:
            - before vs after popularity
- **Social / Viral Trend Data**
    - connects attention → adoption
    - <u>Data sources:</u>

        - TikTok trending sounds (unofficial datasets / scrapes ???)
        - Google Trends
        - YouTube trending / view counts
    - Use cases:
        - detect spikes in attention
        - Compare:
            - search interest vs chart presence
        - Identify:
            - “globally viral vs locally contained”
- **Artist Metadata (Origin + Language)**
    - directly ties to cultural diffusion
    - Data sources:
        - MusicBrainz API
        - Last.fm API
        - Wikidata (artist nationality, language)
    - Use cases:
        - Track:
            - artist country of origin
            - language of songs
        - Compare:
            - where artists succeed outside their origin
- **Live Events + Touring Data**
    - Shows cultural adoption beyond listening
    - Data sources:
        - Eventbrite API
        - Ticketmaster API
        - Bandsintown API
    - Use cases:
        - See:
            - where artists perform
            - where demand exists
        - Compare:
            - streaming popularity vs real-world presence
- **Language + Lyrics Data**
    - directly answers: “Are people listening outside their language more?”
    - Data sources:
        - Genius API (lyrics + metadata)

        - Musixmatch API
    - Use cases:
        - Analyze:
            - language usage in popular songs
        - Track:
            - rise of non-English music
- **Audio + “Mood” Clustering**
    - Data sources:
        - Spotify audio features
        - Kaggle Spotify datasets
    - Use cases:
        - group songs by:
            - mood / vibe
        - Compare:
            - regions by sound
- **Wikipedia / Popularity signals**
    - Measures interest outside streaming platforms
    - Data sources:
        - Wikipedia page views API
    - Use cases:
        - Track:
            - spikes in artist attention
        - Compare:
            - regions (if available)

Eli Christmas Comments/Notes 4/4/2026

I have to say, I love how you structured this. I structure my notes so similarly to what you wrote above, so this is wonderfully absorbed by my brain. Thank you for the time you put into it!

I really like the third/last proposal, and am open and willing to talk about moving forward with all proposals if you guys prefer a different one. The UI ideas immediately started flowing, and I think that would be super fun and I would love to personally use it in my own life and I know a lot of my friends would too.

I'm stoked to help make the data come to life for users.
I've also added a list of potential app APIs with links, and images of UI ideas/components I like and don't like and why.

Questions I have for Tuesday:

- *Question 1*: I asked this in discord chat too but for the purpose of documenting maybe I should ask here, too. What are both of your thoughts on using maps in some way? I think that would be super cool, and I'm also okay with it if we don't use maps. If we didn't use maps, would it be like a list/grid of names of places? Totally open to other suggestions.
- *Question 2*: How soon can we decide on a color palette? Like 2-3 colors, and from that we can select 2 background colors, text colors, and detail/area colors.
- *Question 3*: Have we all found a few screenshots of UI's that we love and a few that we hate? I added mine to the end.
- *Question 4*: Are we including the ability to play the music via Spotify's API, or are we including links for users to listen via various platforms, or maybe just using Spotify's preview feature? From a UI perspective, embedding playback might be cleaner than showing multiple platform links for every song, but full Spotify playback would require users to log in, which might add friction that I bet none of us want to do so I bet that's ruled out for totally obvious reasons. Another option could be using Spotify preview clips instead, which wouldn't require login and would still give users a quick

way to hear the song. If we go the link route, we could also automate link generation so we're not manually managing multiple platforms. I'm mostly trying to figure out which approach ends up being simpler overall. It just depends on if we think users being able to hear the song in-app is integral or not.

- *Question 5*: Can we collectively look at the info the project outline has for the timeline for week 1 and 2 and make sure we're achieving it and discuss what needs done?
    - Week 1: Define problem, roles, tech stack, success metrics. Gather datasets.
        - I think we've defined our problem
        - Roles are clear
        - Did we decide on tech stack? I know it was spoken about, and we discussed react native
        - Success metrics need to be discussed
        - We're in the process of gathering datasets/data. I found a few sources for datasets that I included below
    - Week 2: Data schema, model architecture, UI wireframes.
        - Data schema: things like deciding what our data looks like. For example
            - Countries we'll include
            - Sources (Spotify, Last.fm, etc.)
            - Dates we'll cover of music, like are we only including music created past a certain decade to help datasets not be unrealistically large
            - Defining: "what fields do we store/pull?"
        - Model architecture: the logic for how our app calculates things (is this the right interpretation of model architecture do you guys think?)

        - Define how hidden songs are found
        - Define how regions are compared
        - Define what counts as “discovery”
    - UI wireframe: what will the app look like, a rough draft layout
        - Totally just throwing out ideas to get ideas flowing, not suggesting anything in particular:
        - Map screen
        - Comparison screen
        - ‘Hidden’ music screen
        - Genre screen

Three Potential Map API’s:

---

1. Globe.gl:
    - Simple 3D spinning globe library made for visualizing data between locations
    - Can rotate the globe, zoom in, click countries or cities, add markers, and draw lines connecting regions
    - Completely free and doesn’t require an API key, makes it easy to use for our capstone.
    - Main strength is showing relationships, like music spreading between places, but the default look is more technical and less polished unless you customize it.
    - Oh no I just realized it looks like its web only, I’ll add a third option
    - https://globe.gl/
2. Mapbox Globe:
    - More polished interactive globe built on real map data with smooth zooming, spinning, and clickable regions
    - Can highlight countries, add markers, and animate movement while also customizing the visual style to match our app UI's color schemes/aesthetic, whatever that may end up being

- Requires a free API key, but the free tier caps at an insanely high amount of users so that should be totally okay for what we're doing, and even bigger than what we're doing
- Compared to Globe.gl, it looks more refined and product-like while still supporting the same kinds of interactions
- Web and mobile friendly, and there's a dedicated library called React Native Mapbox
- https://www.mapbox.com/blog/globe-view

3. MapLibre:
    - Free, open-source interactive map library that works on both web and React Native mobile
    - Can zoom around the world, click countries or regions, add markers, draw connections, and fully customize the map style
    - Completely free with no required API key, and you can use free map tile providers or host your own if needed
    - Very similar capabilities to Mapbox, but without usage limits, making it a safe long-term option for a capstone
    - Supports both web (MapLibre GL JS) and mobile using React Native MapLibre,
    - Doesn't have a built-in spinning 3D globe like Mapbox, and is less cool looking, but totally still supports clean world maps and interactive region selection
    - https://maplibre.org/

Potential Data APIs to Use:

Note: *I went through the list of data source ideas and I love them, and thought we could link some API's here for the ones mentioned above. Below these are some additional ones I think could be useful as well.*

- For movie & TV soundtrack data:
    - (I didn't see a free official API for IMDb itself (correct me if I'm wrong of course please) but these pull IMDb-style soundtrack/credits data:
    - TMDB (The Movie Database)
        - https://developer.themoviedb.org/docs
        - Movies, shows, release dates, cast, and sometimes soundtrack references. Commonly used and free with API key.
    - OMDb API (IMDb-linked metadata)
        - https://www.omdbapi.com/
        - This pulls IMDb-based movie data. Not soundtrack-heavy, but could be useful if we needed to link movies to music data.
- For Spotify soundtrack / playlist / song data:

    - Spotify Web API (official and free tier)
        - https://developer.spotify.com/documentation/web-api

- For TMDB (movie + show metadata):
    - The Movie Database:
        - https://developer.themoviedb.org/reference/intro/getting-started

- For Billboard charts (including soundtracks):
    - There's an unofficial but widely used Billboard API wrapper:
        - https://github.com/guoguo12/billboard-charts
    - or npm version:
        - https://www.npmjs.com/package/billboard-top-100

- For TikTok trending songs/sounds (TikTok doesn't have an official public API from what I can see, but these exist):
    - Unofficial TikTok trending API:
        - https://rapidapi.com/Lundehund/api/tiktok-api23
        - (free tier available)
    - TikTok trending scraper datasets:
        - https://github.com/drawrowfly/tiktok-scraper
        - These can pull:
            - Trending sounds
            - Trending videos
            - Music used in videos
            - Hashtags
            - Engagement metrics

- Also: alternative for TikTok signal (very reliable)
    - Google Trends API:
    - https://developers.google.com/search/blog/2025/07/trends-api
    - We can track:
        - Artist name spikes
        - Song name spikes

        - Region interest
        - Timeline trends
        - Often cleaner than TikTok scraping, from what I can see

- For Artist Metadata (Origin + Language):
    - Genius API:
        - https://docs.genius.com/
        - We can get:
            - Lyrics metadata
            - Artist info
            - Song info
            - Language indicators
    - Musixmatch API:
        - https://developer.musixmatch.com/
        - We can get:
            - Lyrics
            - Language detection
            - Song metadata

- For Audio + "Mood" Clustering:
    - Spotify Audio Features API:
        - https://developer.spotify.com/documentation/web-api/reference/get-audio-features
        - We can get:
            - Danceability
            - Energy
            - Valence
            - Tempo
            - Acousticness
            - mood-like data
            - Perfect for vibe clustering and regional sound comparison.
    - Kaggle Spotify Datasets:
        - https://www.kaggle.com/search?q=spotify%20in%3Adatasets

            - Prebuilt datasets including:
                - Audio features
                - Genres
                - Popularity
                - Regions

- For Wikipedia/Popularity Signals:
    - Wikipedia Pageviews API:
        - https://doc.wikimedia.org/generated-data-platform/aqs/analytics-api/
        - We can get:
            - Page view counts
            - Spikes in attention
            - Timeline popularity

Note: *Here are some additional API's that I found that I think aligns with our question and purpose of the project.*

- For radio airplay / regional exposure:
    - Purpose: This helps answer: "what's actually being played locally?"
    - Radio Browser API:
        - https://api.radio-browser.info/
        - We can get:
            - Radio stations by country
            - Genres per region
            - Station metadata
            - Streaming URLs
        - Useful for:
            - Compare streaming vs radio popularity
            - See regional sound differences
            - Detect local-only artists

- For YouTube music popularity by region:
    - Purpose: This helps detect songs trending without Spotify.
    - YouTube Data API:
        - https://developers.google.com/youtube/v3
        - We can get:
            - Trending videos by region
            - Music videos popularity
            - View counts
            - Publish dates
        - Useful for:
            - Viral before Spotify charts
            - Regional video popularity
            - Cross-platform discovery

- For Apple music charts (regional popularity):
    - Purpose: Spotify-only data can bias results, so this adds another platform.
    - Apple Music Charts (unofficial but widely used):
        - https://github.com/kylehowells/apple-music-api
        - Or https://rss.applemarketingtools.com/
        - We can get:
            - Top songs by country
            - Top albums by region
            - Chart rankings
        - Useful for:
            - Compare Spotify vs Apple audiences
            - Detect platform-specific trends

- For popularity (what people are discovering):
    - Purpose: This is really cool for the "discovery gap"
    - Last.fm API:
        - https://www.last.fm/api
        - We can get:
            - Trending tracks

                - Listener counts
                - Popularity growth
                - Global discovery
            - Useful for:
                - Detect emerging artists
                - Track discovery spikes
                - Compare discovery vs charts
        - We can even use it geographically:
            - https://www.last.fm/api/show/geo.getTopTracks
            - We can get:
                - Top tracks by country
                - Regional popularity
                - Artist discovery by location
                - Listener counts

- For geo/country metadata (for UI maps):
    - Purpose: Helps normalize country data across APIs.
    - REST Countries API:
        - https://restcountries.com/
        - We can get:
            - Country names
            - Codes
            - Regions
            - Coordinates
        - Useful for:
            - Map plotting
            - Country normalization
            - Region grouping

- For language detection (automatic):
    - Purpose: Instead of manually tagging language. But I think language is probably in the metadata we already have access to, this was just worth mentioning too.
    - LibreTranslate API (free):

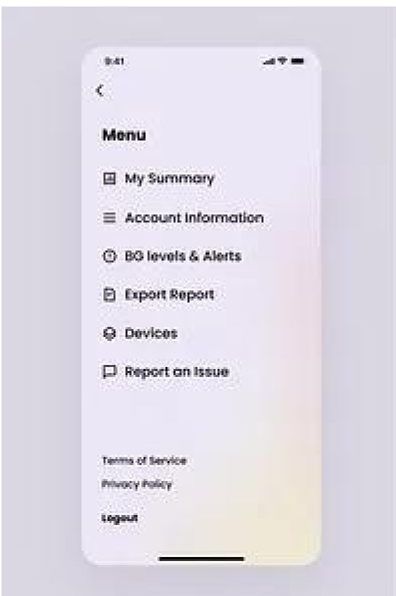
        - https://libretranslate.com/
        - We can:
        - Detect song language
        - Translate titles
        - Group by language
    - Useful for:
    - Non-English music tracking
    - Cross-language spread

- For genre taxonomy/classification:
    - Purpose: Helps normalize genres across APIs.
    - Every Noise at Once dataset:
        - https://everynoise.com
        - Or https://github.com/everynoise
        - We can:
            - Group genres
            - Map genre relationships
            - Cluster sounds
        - Useful for:
            - Regional sound clustering
            - Genre diffusion

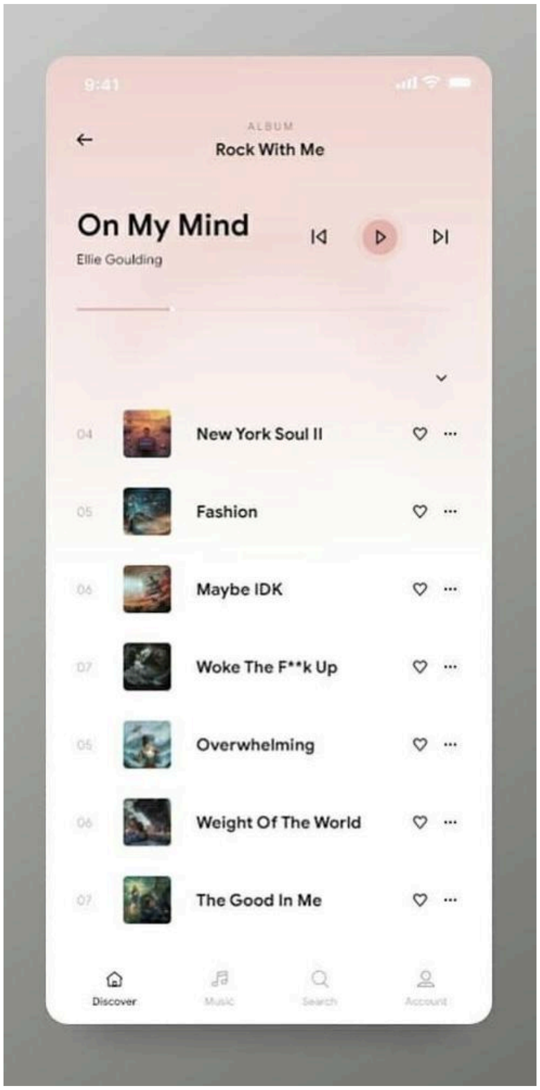
- For social media popularity signals (non-TikTok):
    - Reddit API:
        - https://developers.reddit.com/docs/capabilities/server/reddit-api
        - We can:
            - Track artist mentions
            - Subreddit popularity
            - Regional communities
        - Useful for:
            - Showing discovery
            - Niche artist growth

App UI images I like and don’t like and why:

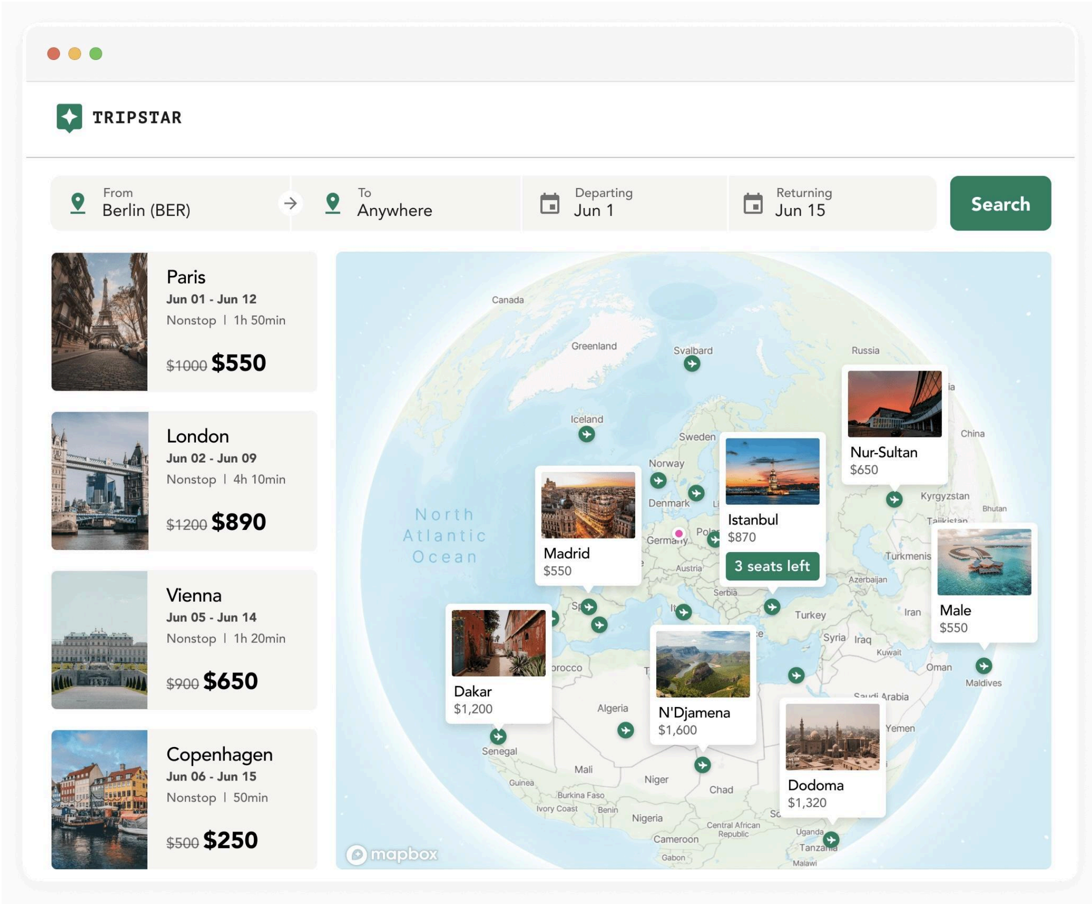
I like the images below for very specific reasons and explain why. Also please keep in mind this means nothing at all about the UI images shown as a whole. I specify what I like about each. So unless it says so, I don’t mean I like the specific font or colors or whatnot, if that makes sense.



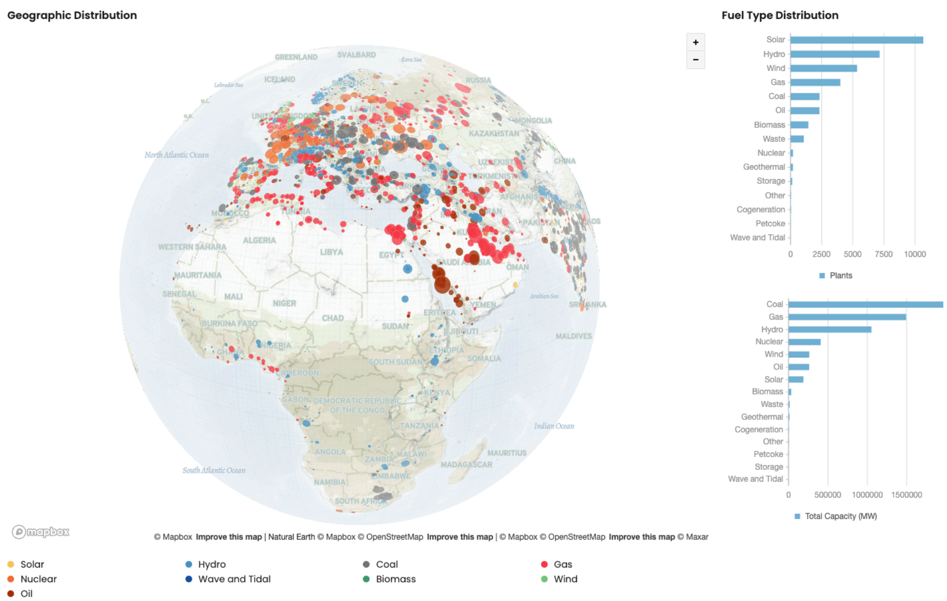
I like this as a general menu/list concept because it’s simple and accessible. I like the gradient. The gradient background is easy to make and easy to shift colors slightly on different screens for variation and cohesiveness. It would be easy to have a light and dark mode for this concept.



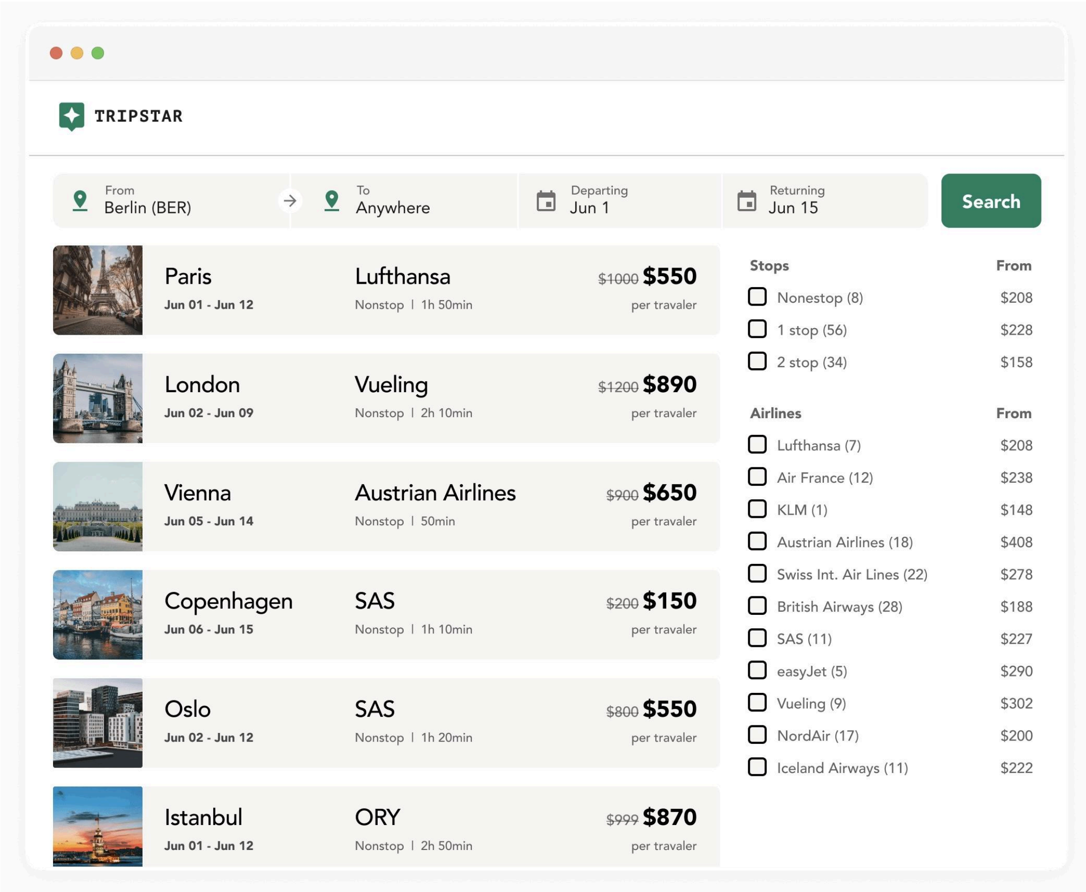
I like this as a general list screen UI concept because it's simple and easy to mentally absorb. The brain can easily see everything and know all available buttons. I like the gradient. It would be easy to have a light and dark mode for this concept.



I like this as a general combined globe/map and list screen concept because it shows filtered suggestions in both ways. It could be filtered to a specific genre of music for example and then users can move the globe (and/or see the list on the side) to explore. It uses one of the map AI’s, but can generally be achieved using all.



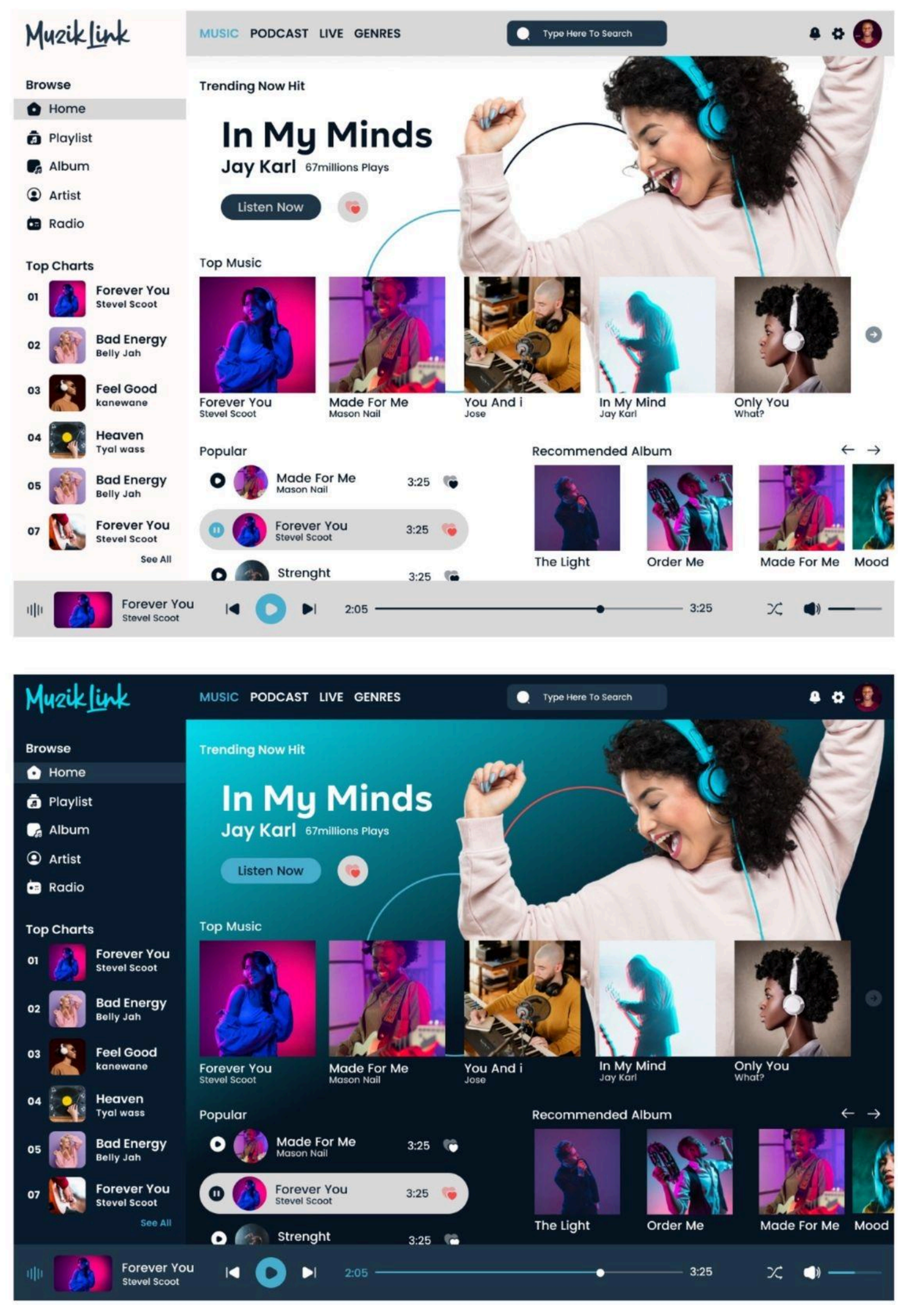
I like this as another general combined globe/map and info screen UI concept because it shows filtered suggestions in both ways. Same with the above image, it could be filtered to show a specific genre or artist and then users can move the globe (and/or see the list on the side) to explore.



I like this as a list view UI concept because of the same reasons above but this is specifically list view.

I don't like these:

I don't like this because it's busy, and looks like the bubbles probably move. It's a rad idea, but it's not very accessible.



I don’t like this because it’s so busy, it’s a lot of mental load immediately and feels like there’s so much in the app it’s hard to discover it all. And if accessible fonts were applied it would be even busier. I think we need to be mindful of the brightness and colors of the album art.